

Supremum and Infimum

Benjamin Abshire[†]
Kent State University

September 30, 2025

I Introduction

To understand what supremum and infimum are, we first must define when a set is bounded. Let S be a non-empty subset of $\mathbb{R}$. Then,

- i. The set S is *bounded above* if there exists a number $u \in \mathbb{R}$ such that $s \leq u$ for all $s \in S$.

Each u is called an *upper bound* of S .

- ii. The set S is *bounded below* if there exists a number $w \in \mathbb{R}$ such that $w \leq s$ for all $s \in S$.

Each w is called an *lower bound* of S .

- iii. The set S is *bounded* if it is bounded both above and below.

A set is *unbounded* if it is not bounded

If a set has one upper bound, then it has infinitely many upper bounds, as you can just add 1 to an upper bound to find the next. If a set has one lower bound, then it has infinitely many lower bounds, for similar reasons.

II Definitions and Theorems

The *supremum* of a set is the least upper bound of that set. There may be many upper bounds, but the supremum is less than all of them. Formally, let S be a non-empty subset of $\mathbb{R}$. If S is bounded above, then a number u is the supremum of S if:

- i. u is an upper bound of S
- ii. if v is any upper bound of S , then $u \leq v$

The *infimum* of a set is the greatest lower bound of that set. There are many lower bounds, but the infimum is greater than all of them. Formally, let S be a non-empty subset of $\mathbb{R}$. If S is bounded below, then a number w is the infimum of S if:

- i. w is a lower bound of S
- ii. if t is any lower bound of S , then $t \leq w$

[†]Definitions and example problems from MATH 42001: Analysis I instructed by Dr. Artem Zvavitch and *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert

Alternative Definitions

We can re write part *i* of the supremum definition as the following equivalent statement:

$$s \leq u, \forall s \in S$$

And we can rewrite part *ii* of the supremum definition as the following three equivalent statements:

if $z < u$, then z is not an upper bound of S

if $z < u$, then there exists s_z such that $z < s_z$

if $\epsilon > 0$, then there exists $S_\epsilon \in S$ such that $u - \epsilon < s'$

Therefore, we can also say that a number u is the supremum of a nonempty subset S of $\mathbb{R}$ if and only if u satisfies:

i. $s \leq u, \forall s \in S$

ii. if $v < u$, then $\exists s' \in S$ s.t. $v < s'$

Finally, we can also say that an upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if:

$$\forall \epsilon > 0, \exists s_\epsilon \in S \text{ s.t. } u - \epsilon < s_\epsilon$$

Functions

Consider a function $f : D \rightarrow \mathbb{R}$. f is bounded above if the set $f(D) = \{f(x) : x \in D\}$ is bounded above in $\mathbb{R}$; that is:

$$\exists B \in \mathbb{R} \text{ s.t. } f(x) \leq B \forall x \in D$$

f is bounded below if the set $f(D)$ is bounded below. f is bounded if it is bounded above and below; that is:

$$\exists B \in \mathbb{R} \text{ s.t. } |f(x)| \leq B \forall x \in D$$

The Completeness Property of $\mathbb{R}$

Every nonempty set of $\mathbb{R}$ that has an upper bound also has a supremum in $\mathbb{R}$

The Archimedean Property

If $x \in \mathbb{R}$, then $\exists n_x \in \mathbb{N}$ such that $x < n_x$

Proof. If the property is false, then $n \leq x \forall n \in \mathbb{N}$, so x is an upper bound of $\mathbb{N}$. So, by the Completeness Property, the nonempty set has a supremum $u \in \mathbb{R}$. If we subtract 1 from u , we get a number that is smaller than the supremum of $\mathbb{N}$ (since $u - 1 < u$). So, $u - 1$ is not an upper bound of $\mathbb{N}$, so $\exists m \in \mathbb{N}$ with $u - 1 < m$. Adding 1 gives $u < m + 1$ which contradicts that u is an upper bound of $\mathbb{N}$ (since $m + 1 \in \mathbb{N}$). $\square$

Before moving on, note that the supremum of S can be notated by $\sup(S)$ and the infimum of S as $\inf(S)$. Also, $\sup_{x \in D} f(x) = \sup(D)$ and $\inf_{x \in D} f(x) = \inf(D)$.

III Problems

1. Consider a non-empty set $E \subset \mathbb{R}$. Prove that E has a supremum if and only if $-E = \{x \in \mathbb{R} : -x \in E\}$ has an infimum, in which case, $\inf(-E) = -\sup(E)$.

Proof. Let u be an upper bound of E s.t. $s \leq u \forall s \in E$. By the Completeness Property of $\mathbb{R}$, E has a supremum. $-s \geq -u \forall s \in E$, so $-u \leq -E$. Therefore, $-u$ is a lower bound of $-E$. Also, by the Completeness Property of $\mathbb{R}$, $-u$ is the infimum of E .

Since $\sup(E)$ is the upper bound of E , $-\sup(E)$ is the lower bound of $-E$. If $z \in \mathbb{R}$ is a lower bound of $-E$, then $z \leq -s \forall s \in E$ so $s < -z \forall s \in E$. So, $-z$ is an upper bound of E . By the definition of supremum, $\sup(E) \leq -z$ so $-z \leq -\sup(E)$ and $-\sup(E) \geq z$. $-\sup(E)$ is greater than or equal to any lower bound of $-E$, therefore, $\inf(-E) = -\sup(E)$. $\square$

2. Suppose f and g are real valued functions with common domain $D \subset \mathbb{R}$. Determine if the following are true. If so, prove it, if not, provide a counter example.

$$\text{a. } \sup_{x \in D} f(x) + \sup_{x \in D} g(x) = \sup_{x \in D} (f(x) + g(x))$$

Solution. Let $f(x) = \begin{cases} -1 & \text{if } x \neq -1 \\ 1 & \text{if } x = -1 \end{cases}$ and $g(x) = \begin{cases} -1 & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$

$$\sup_{x \in D} (f(x) + g(x)) = -1 + 1 = 0 \neq \sup_{x \in D} f(x) + \sup_{x \in D} g(x) = 1 + 1 = 2$$

$$\text{b. } \sup_{x \in D} f(x) \cdot \sup_{x \in D} g(x) \leq \sup_{x \in D} (f(x) \cdot g(x))$$

Solution. Let $f(x) = \begin{cases} -1 & \text{if } x \neq -1 \\ 1 & \text{if } x = -1 \end{cases}$ and $g(x) = \begin{cases} -1 & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$

$$\sup_{x \in D} (f(x) \cdot g(x)) = -1 \cdot 1 = -1 \neq \sup_{x \in D} f(x) + \sup_{x \in D} g(x) = 1 \cdot 1 = 1$$

3. Find and prove the supremum and infimum of $S = \left\{ \frac{1}{n^2} : n \in \mathbb{N} \right\}$

Proof.

i. The supremum of S is 1.

ii. The infimum of S is 0.

Proof of i. To prove that the supremum is 1 we must show that 1 is an upper bound of S and that it is the least upper bound of S .

$n \geq 1 \forall n \in \mathbb{N} \implies n^2 \geq 1^2 \implies 1 \geq \frac{1}{n^2} \forall n \in \mathbb{N}$. Therefore, 1 is an upper bound of S and $S \neq \emptyset$ and S is bounded above.

Let U be the supremum of S , and let $\epsilon > 0$. $1 \in S$, and $1 - \epsilon < 1$. So, $1 - \epsilon$ is not an upper bound. Therefore $1 = U = \sup S$. $\square$

Proof of ii. $0 \leq \frac{1}{n^2} \forall n \in \mathbb{N} \implies 0 \leq 1$. Therefore 0 is a lower bound of S and $S \neq \emptyset$ and S is bounded below.

Let $w = \inf S$, and let $\epsilon > 0$. By the Archimedean Property, $\exists n \in \mathbb{N}$ such that $\frac{1}{n} < \sqrt{\epsilon}$. So, $0 \leq w \leq \frac{1}{n^2} < \epsilon$. By Theorem 2.1.9, (If $a \in \mathbb{R}$ is such that $0 \leq a < \epsilon \forall \epsilon > 0$, then $a = 0$) since $0 \leq w < \epsilon$ and $\epsilon > 0$ is arbitrary, $0 = w = \inf S$. $\square$

4. Find and prove the supremum and infimum of $(-1, 1)$.

Proof.

i. The supremum is 1.

ii. The infimum is -1 .

Proof of i. 1 is an upper bound, because for $\forall x \in (-1, 1) : x < 1$.

1 is the least upper bound. Assume u is another upper bound and is less than 1. Consider $x = \frac{u+1}{2}$, $x \in (-1, 1)$. Then, $-1 < \frac{u+1}{2} < \frac{1+1}{2} = 1$ and $\frac{u+1}{2} > \frac{u+u}{2} = u$. So, $x \in (-1, 1)$, $x > u$ contradiction. u is not an upper bound. Note that we get $\frac{u+1}{2} < \frac{1+1}{2} = 1$ by $u < 1$. Therefore, by the definition of supremum, 1 is the supremum of $(-1, 1)$. $\square$

Proof of ii. -1 is a lower bound, because for $\forall x \in (-1, 1) : x > -1$.

-1 is the greatest lower bound. Assume v is another lower bound and is greater than -1 . Consider $x = \frac{v-1}{2}$, $x \in (-1, 1)$. Then, $-1 = \frac{-1-1}{2} < \frac{v-1}{2} < 1$ and $\frac{v-1}{2} < \frac{v+v}{2} = v$. So, $x \in (-1, 1)$, $x < v$ contradiction. v is not a lower bound. Therefore, by the definition of infimum, -1 is the infimum of $(-1, 1)$. $\square$

5. Prove that

$$\sup_{x \in D} |f(x) + g(x)| \leq \sup_{x \in D} |f(x)| + \sup_{x \in D} |g(x)|$$

Also, show an example of the two functions where the equality holds and the strict inequality holds.

Proof. By the triangle inequality, $|f(x) + g(x)| \leq |f(x)| + |g(x)|$. By the definition of supremum, $|f(x)| \leq \sup_{x \in D}$ and $|g(x)| \leq \sup_{x \in D}$. So, $|f(x)| + |g(x)| \leq \sup_{x \in D} |f(x)| + \sup_{x \in D} |g(x)|$ and $|f(x) + g(x)| \leq \sup_{x \in D} |f(x) + g(x)|$. Since $|f(x) + g(x)| \leq |f(x)| + |g(x)|$, then, $\sup_{x \in D} |f(x) + g(x)| \leq \sup_{x \in D} |f(x)| + \sup_{x \in D} |g(x)|$. $\square$